

0/20 Questions Answered

hw7-diff-props-single-mean-electric-vehicle-adoption

Student Name

Search students by name or email...



Q1

0 Points

Note: There are three options for homework 7, that differ by data topic, and you only need to complete one of the three assignments.

Electric vehicles offer an alternative transportation method which has a smaller environmental impact than traditional vehicles running on gas, because they have reduced greenhouse gas emissions. Despite this, there is not wide adoption of electric vehicles across the United States. There are many studies which try to identify characteristics related to people's car preference. Egbue, Long, and Samaranayake ([2017](#)) studied the adoption of electric vehicles (EVs) in the United States by professional engineers. They did a Facebook poll in a Facebook group for professional engineers.

	Graduate Degree	No Graduate Degree
Would purchase an electric vehicle in next 10 years	8	25
Would not purchase an electric vehicle in next 10 years	29	51

Consider the above table which depicts the number of engineering professionals polled that specified they would or wouldn't purchase an electric vehicle in the next 10 years, separated by whether or not they have a graduate degree (masters or PhD).

Let

- p_1 represent the proportion of professional engineers with some graduate degree (i.e. masters or PhD) that would purchase an electric vehicle in the next 10 years in this sample
- p_2 represent the proportion of professional engineers with no graduate degree (i.e. only bachelors or below) that would purchase an electric vehicle in the next 10 years in this sample
- π_1 represent the corresponding population parameter for professional engineers with some graduate degree.

- π_2 represent the corresponding population parameter for professional engineers without a graduate degree.

Save Answer

Q2**5 Points**

We want to calculate a 90% CI for the difference in proportion of engineering professionals that would buy an electric vehicle in the next 10 years between those with and without a graduate degree (i.e. $\pi_1 - \pi_2$).

Q2.1**1 Point**

What is the point estimate for the difference in proportions? Enter only your final answer. Round to four decimal places.

[Save Answer](#)**Q2.2****1 Point**

What is the critical value for the 90% confidence interval? Enter only your final answer. Round to four decimal places. Include a picture of your work.

 Please select file(s)[Select file\(s\)](#)[Save Answer](#)

Q2.3**1 Point**

What is the standard error for the confidence interval? Include a picture of your work, including the formula you used.



 Please select file(s)

Select file(s)

Save Answer

Q2.4**1 Point**

With the point estimate, critical value, and standard error we estimate the 90% CI for the difference in proportion of engineering professionals that would buy an electric vehicle in the next ten year, between those with and without graduate degrees, to be $(-0.2550, 0.0296)$.

We will change the question slightly. Now we want to calculate a 90% CI for the difference in proportion of engineering professionals that would buy an electric vehicle in the next ten year, between those without and with graduate degrees, to be (i.e. $\pi_2 - \pi_1$). Note the order of the groups has changed now.

Think about which parts of the confidence interval formula change when you change the order of the difference in proportions.

The lower bound of the new confidence interval is: (Enter only your final answer and round to four decimal places)

The upper bound of the new confidence interval is: (Enter only your final answer and round to four decimal places)

Save Answer

Q2.5**1 Point**

What do you notice about the relationship between the two confidence intervals we get when changing the order of the groups?

Save Answer

Q3**6 Points**

Now we want to conduct a hypothesis test to test if there is evidence of a difference in proportions, with a significance level of 0.1.

$$H_0 : \pi_1 = \pi_2$$

$$H_A : \pi_1 \neq \pi_2$$

Q3.1**1 Point**

Calculate the pooled proportion assuming the null hypotheses is true ($H_0 : \pi_1 = \pi_2$). Enter only your final answer and round to four decimal places. Include a picture of your work.

 Please select file(s)**Select file(s)****Save Answer****Q3.2****1 Point**

Calculate the standard error for the pooled proportion. Include a picture of your work including the formula used.

 Please select file(s)**Select file(s)****Save Answer**

Q3.3**1 Point**

If the null hypothesis were true, what would be the sampling distribution of the difference in sample proportions?

[Save Answer](#)**Q3.4****1 Point**

Now we need to calculate the p-value so we can judge if there is a statistically significant difference in proportions. Enter only your final answer and round to four decimal places. Then upload a picture of your work.

 Please select file(s)[Select file\(s\)](#)[Save Answer](#)

Q3.5**2 Points**

You should have found a p-value that is more than 0.1. Select the following option that is an appropriate conclusion, at a significance level of 0.1, from the results of our hypothesis test.

We accept the null hypothesis and conclude there is no evidence of a difference in proportion of engineering professional that would buy an electric vehicles in the next 10 years between those with and without a graduate degree.

We fail to reject the null hypothesis and conclude there is no evidence of a difference in proportion of engineering professional that would buy an electric vehicles in the next 10 years between those with and without a graduate degree.

We fail to reject the null hypothesis and conclude there is evidence of a difference in proportion of engineering professional that would buy an electric vehicles in the next 10 years between those with and without a graduate degree.

We reject the null hypothesis and conclude there is evidence of a difference in proportion of engineering professional that would buy an electric vehicles in the next 10 years between those with and without a graduate degree.

We reject the null hypothesis and conclude there is no evidence of a difference in proportion of engineering professional that would buy an electric vehicles in the next 10 years between those with and without a graduate degree.

[Save Answer](#)

Q4**2 Points**

We forgot to check that the assumptions were met for the confidence interval and hypothesis test we performed! Please identify the conditions and explain if they were met.

[Save Answer](#)

Q5**8 Points**

Consider the following hypothetical situation:

A certain model of electric vehicles is reported to be able to go 300 miles (consider this as the reported population mean) on one charge. A researcher wants to know if, in fact, the average distance is correctly reported, with a significance level of 0.05. She test drives 100 electric vehicles of that model and gets a sample mean of 275 miles and a sample standard deviation of 25 miles.

Q5.1**1 Point**

Consider the hypothesis testing procedure.

The null and alternative hypothesis are:

$$H_0 : \mu = 300$$

$$H_A : \mu \neq 300$$

What is the point estimate? Enter only your final answer.

[Save Answer](#)

Q5.2**1 Point**

What is the standard error of the point estimate? Enter only your final answer. Include a picture of your work

 Please select file(s)**Select file(s)****Save Answer****Q5.3****1 Point**

If the null hypothesis were true, what is the sampling distribution?

Save Answer**Q5.4****2 Points**

Now we need to calculate the p-value so we can judge if there is evidence of the average an electric vehicle could travel on a single charge is different from 300. Enter only your final answer and round to four significant digits. Then upload a picture of your work.

 Please select file(s)**Select file(s)****Save Answer**

Q5.5**2 Points**

You should have calculated a p-value much smaller than the significance level, 0.05. Explain to your friend Lukas, who is considering getting that model of electric vehicle, what this p-value means, in context. Tell him if there is evidence, based on this data, that the average amount of miles that model of electric vehicle can travel on a single charge is different from 300 miles.

[Save Answer](#)**Q5.6****1 Point**

Say the researcher then takes another sample. With the new data of 100 cars the sample mean is still 275 miles, but the sample standard deviation this time is 250. State how the p-value would change and explain why this change makes sense.

[Save Answer](#)

Q6**0 Points**

The following two questions are related to the homework choice you made in choosing to do this assignment instead of the other two option. There are no correct answers, this question is only being graded for completeness (note that answering "I felt like it" is an insufficient response).

Please explain in detail what led you to choose this topic, sleep deprivation, to investigate.

What was it about the other two options (sleep deprivation and cheating pre- and post- AI) that led you not to choose them? Please explain in detail.

[Save Answer](#)

Q7**0 Points**

- Have you done the homework by yourself, with a peer(s), or used tools like ChatGPT? If you do not acknowledge any help from a person or ChatGPT, we will consider this cheating.

- If you provided help to a peer(s) and/or got help from your peers(s) please explain briefly what this entailed.

- If you used ChatGPT, include the prompts you used and explain how ChatGPT supported your learning.

[Save Answer](#)[Save All Answers](#)[Submit & View Submission >](#)